

Monitoring of Land Cover and Land-Use Changes Using Geospatial Techniques in Pulivendla Mandal, A.P.

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Abstract:

The study of land use/land cover (LU/LC) variations is critical for proper planning, usage, and management of natural resources. Traditional methods for gathering demographic data, censuses, and environmental sample analysis are insufficient for multicomplex environmental studies due to the numerous problems that frequently arise in environmental issues and the high complexity of handling multidisciplinary data sets; new technologies such as satellite remote sensing and Geographical Information Systems are required (GIS). Our findings show that LU/LC fluctuations were significant from 2007 to 2018. The built-up area has increased dramatically. There has been a huge increase in the built-up area. On the other side, agricultural land, water spread land, and forest lands are all dwindling. The impact of population and development activities on LU/LC change is clearly demonstrated in this study. This research demonstrates that combining GIS with remote sensing technology might be a useful tool for urban planning and management. This project will help environmental management organizations, legislators, and the general public better comprehend their surroundings.

Keywords: Land use, land cover, Pulivendula, Remote Sensing, GIS.

1. Introduction:

Land is both a resource and a resource base in and of itself. It is the physical environment in which man lives

and interacts. Vast stretches of water cover a large portion of the earth's surface. Land area accounts for only 29% of the total surface area of the planet (Burrough et al., 1996; Chauhan, 2003). As a result, man must rely on less than one-third of the surface as a habitable area. More over two-thirds of this land surface is judged to be unsuitable for human habitation. Although the phrases land use and land cover is sometimes used interchangeably, each has its own distinct meaning. Land cover—defines the qualities and surface cover of the Earth's surface as represented by natural elements such as vegetation, water, bare earth, impervious surface, and other physico-chemical properties. Land cover identification provides the foundation for tasks such as thematic mapping and change detection analysis. Land use refers to the activity, economic purpose, planned use, and/or management plan those humans or land managers have placed on the land cover type(s). Land use change can also be defined as a shift in intention or management practice. The terms Land Use / Land Cover, when used interchangeably, refer to the categorization or classification of human activities and natural elements on the landscape through time using accepted scientific and statistical methods of analysis of acceptable source materials (Houghton, 1994; Kaul et al., 2009; Mohammed, 1993). The physical stuff that covers the earth's surface is known as land cover. Land use is a term that describes how people use land for socioeconomic purposes; urban and agricultural land uses are two of the most well-known high-level classifications of use. There may be several and alternative land uses at any given site or location, each with its own set of specifications. As a result, land use refers to the human activity of using land. Thus, data available on LULC changes can offer critical input to decision-making of environmental management and planning the future (Sunitha et al., 2013; 2014; Prasad et al., 2014; 2019).

Remote sensing data and techniques, as well as Geographical Information Systems (GIS), provide effective methods and tools for analyzing land use and land cover characteristics. Planning and modeling of land cover. V. Sunitha et al., (2011) investigated remote sensing analysis

of land resources for planning and development in and around the pulivendula- sanivaripalli area of Andhra Pradesh, concluding that remote sensing data give trustworthy, accurate baseline information for land use / land cover mapping. Sharma et al., (2005) used remote sensing and GIS techniques to investigate the impact of coal mining on land use/land cover in the Jaintia hills district of Meghalaya, India. They used LANDSAT data from 1975, 1987, 1999, and 2007 to conclude that there was a fourfold increase in mining area from 1975 to 2007, accompanied by a threefold decrease in forest area. Integration of Remote Sensing data in GIS is very useful in delineating groundwater potential zones in various ways (Sunitha et al., 2013; 2019; Sudharshan et al., 2020).

The most accurate and up-to-date map available for studying land use and land cover, which leads to change detection, is usually satellite remote sensing data. It is the finest method for tracking urban growth/sprawl, especially in India's fast-growing towns and cities (Hellden and Stern, 1980, Sabins, 1987, Frederiksen, 1993, Mohammed, 1993, Raina et al., 1993. Sujatha et al., 2000). For land use and land cover issues, a customizable geographical database can be built using a geographical information system. As a result, this tool is quite useful in presenting the results of the socioeconomic survey (Spatial and Non-Spatial data). For good planning information on all the above factors, knowledge of both land use and land cover is essential. Every parcel of land on the Earth's surface, according to Sharma et al., (2005), is unique in terms of the cover it has. Land use and land cover are two separate but intertwined aspects of the Earth's surface. Grazing, agriculture, urban development, logging, and mining are just few of the uses for which land might be utilized. Cropland, woodland, wetland, pasture, highways, and urban areas are examples of land cover classifications. The city is without a doubt one of the most remarkable manifestations of modern civilization, or perhaps any civilization. It entirely dominates the region in which it is located, and in that sense, the word may be seen as increasingly becoming a collection of towns and cities, as

well as the regions that they serve and rely on. Cities are "both places and the total of the people who live in them," therefore studying them is an essential part of human geography (Burrough, 1986; Sabins, 1987; Chauhan, 2003). In India, a town or city is defined as a municipality, corporation, cantonment, or notified region. As the human population develops, more space is required to accommodate the expanding population and their functions, resulting in the formation of various functional zones, particularly in metropolitan areas. The primary goal of studying the LULC of the study area is to better manage developmental activities such as urban expansion and forest deforestation, as well as to monitor agriculture and other developmental activities (settlement, pasture, and water bodies) and take appropriate remedial measures as needed in the study area.

2. Study Area:

Pulivendula Mandal is an over-exploited mandal in Andhra Pradesh's Kadapa district, which is economically disadvantaged and plagued by drought. It is situated between north latitudes 14.32 and 14.49, and east longitudes 78.12 and 78.32, the mandal are located. The mandal is flanked on the north by Thondur mandal, on the east by Vemula mandal, on the south by Ananthapur district, and on the west by Ananthapur district. It is located in the western section of the Kadapa district. (Fig.1). Geographically, the study included 182.6 square kilometers, with the study region having a dendritic pattern due to geological formations.

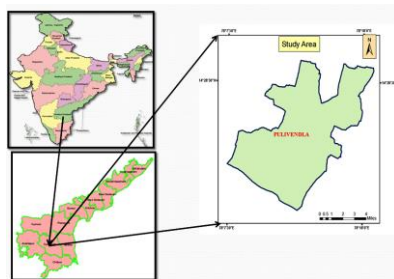


Fig. 1 Location Map of the Study Area

2.1 Geology of the Study Area:

Geologically the study area occupied with Peninsular Gneissic Complex of Archean and Papaghni and Chitravathi group of rocks such as basal polymictic conglomerate/grit, arkose, quartzite with intercalations of shale and overlies the Archean to Paleoproterozoic basement consisting of granite/ gneiss and narrow linear schist belts with a profound Eparchean unconformity (Rao et al., 1987). The arenaceous sequence of Gulcheru quartzite is successively overlain by dominantly calcareous rocks of Vempalli formation having litho units of stromatolitic dolomites at places, shale, mudstone and chert. Basaltic lava flows and dolerite sills/dykes are present in this formation. A number of transverse faults and fractures affecting both the basement and the overlying sediments are basement rooted and subsequently reactivated (Sunitha et al., 2016)

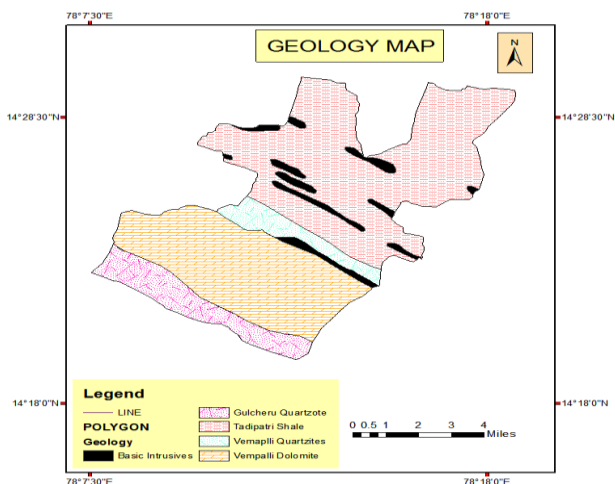


Fig. 2 Location Map of the Study Area

The area is underlain by Meta Sedimentary formations comprising of Shale, Limestone and Quartzite of Pre-Cambrian age. Ground water occurs in weathered

and fractured zones under water table and semi- confined conditions.

3. Methodology:

A Geographic Information System (GIS) database is made up of a series of geographically recorded data planes or item themes that may be mapped. GIS is a graphical representation of data that also includes tools for analyzing geographically referenced data. For the purpose of study of land use / land cover ARC GIS 9.3 and ERDAS IMAGINE 2014 are powerful tools for extracting the land use, land cover layers, from satellite imageries. The land use/land cover classes include agriculture land, settlements, rivers, barren land etc. Agriculture land, settlements, rivers, barren land, and other land use/land cover types are included. Figure 3 depicts the approach used to map the study area's land use and land cover maps. Later Change Detection methodology was done for the images to find out the changes that have taken place in the study region using ERDAS IMAGINE 2014. The feature classifications were determined using a combination of visual interpretation of satellite imagery and field checks. For both years, these datasets were digitized and analyzed to generate land use/land cover statistics for the areas classified under each of these categories. In light of the foregoing, attempts were made to analyze the spatial distribution of Land use and Land cover during the year of 2007 to 2018 and generate maps using GIS techniques while conducting the current study of Land use and Land cover across the Pulivendla mandal. Images were acquired from BHUVAN portal for the study of land use and land cover in the Pulivendla mandal and the results were supervised and classified. Table 1 shows the picture interpretation keys used for land use and land cover classification. After limited field checks; these pre-field thematic maps are changed, substantiated, and confirmed.

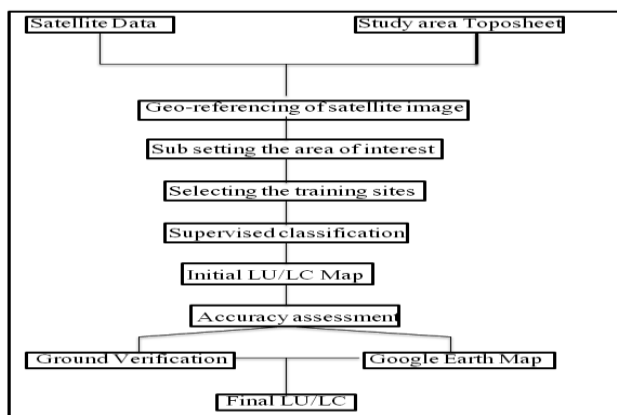


Fig. 3 Flow Chart Showing the Broad Methodology Adopted in the Study

Preparation of Base Map: The base map is a must-have for any mapping project that involves remote sensing or ground-based approaches. It is a map that is used to collect and display thematic (resource) information on a shared standard scale. For base map production in the study area, we used SOI toposheet No. 57 J 07 and J 11 on a 1:50,000 scale. The significant features, such as key towns or villages, rivers, and reserved forest areas, were sketched on a tracing sheet, from which a 1:50,000 scale base maps were created.

Preparation of Thematic Maps: These maps accurately depict earth phenomena such as the geographic distribution of natural resources at the time of survey (Gupta, 2003). In this study, a satellite picture (IRS-P6 LISS-III) was used as the base map, which is a true record of the diverse environmental resource's information. Visual interpretation of satellite imagery is used to create these maps depicting the geographic distribution of forest, agricultural, and water resources, among other things. Visual interpretation is based on visual properties such as tone, size, shape, pattern, texture, and others, as well as existing maps and literature.

Land use class		Image characteristics	Tone					
			Band 5	Band 7	Band 4,5&7	Texture	Pattern	Size
Built up land		Lighter	Medium to light	Blue mixed with grey	Fine red and white	Semi circular to circular	Small	In plains plateaus along the rivers, roads and railways
Agricultural Land	Crop land	Dark grey	Light to medium	Red	Fine	Distinct	Vary in size	Plains and vally land
		Fallow land	Light grey	Light grey	Yellowish	Rough	Irregular	
Forest land	Fairly dense	Very dark grey	Light to medium	Light red to brown	Fine to medium	Irregular with distinct	Varying Irregular size	High land hills
Water bodies	River/stream	Sparse	Medium dark grey	Light grey	Light brown	Rough	Irregular	Natural stream
		Medium to dark grey	Dark grey	Medium blue to dark	Fine	Linear	Long	
	Lake/tank	Dark grey	Very dark grey	Light to dark blue	Fine	Triangular	Medium	Surrounded by hills
Uncultivable land	Barren land	White grey	Light grey	White to yellowish	Fine	Distinct	Large	Hilly terrain along
	Waste land	Light to medium	Medium grey	Brown to grey	Fine	Distinct	Small	Rivers/ streams

Table. 1 Image Interpretation Keys Used for Land Use/ Land Cover Classification

4. Results and Discussions:

A total of three thematic maps, including land use/land cover, drainage, and lineament maps, were created based on image interpretation and limited field inspections to examine the various land resources in the research area.

4.1 Category Wise Land Use Land Covers Dynamics:

Built up Land: That area was investigated between 2007 and 2018, the amount of built-up land has decreased significantly (Fig. 4). Built-up land covered an area of 19.39 sq km in 2007, accounting for 10.68 percent of the total

area, compared to 4.935 sq km in 2018, accounting for 2.71 percent of the total area (Fig. 5). The amount of built-up land has reduced during the last two decades, possibly at the expense of forest cover.

Scrub Land: It was discovered that the area under scrub land has decreased from 2007 to 2018 (Fig. 4). Built-up land covered 82.63 square kilometers in 2007, accounting for 82.63 percent of the total area, compared to 63.9 square kilometers in 2018, accounting for 45.51 percent of the total area (Fig. 5). Scrub land acreage has reduced in the last two decades from 2018 to 2007, maybe at the expense of forest cover.

Fallow Land: It was discovered that between 2007 and 2018, the area of fallow land increased significantly (Fig: 4). Fallow land covered 27.74 sq km in 2007, accounting for 15.28 percent of total area, rising to 60.15 sq km in 2018, accounting for 32.12 percent of total area (Fig: 5). The area of fallow land has expanded over the last two decades, possibly at the expense of forest cover.

Water Body: It was discovered that the area's water body had increased in size from 2007 to 2018. (Fig: 4). In 2007, a water body covered 0.011 square kilometers, accounting for 0.0064 percent of the total area, however in 2018, it covered 2.13 square kilometers, accounting for 1.17 percent of the entire area (Fig: 5). The increase in water body area from 2007 to 2018 could be attributed to the discharge of water from the Gandikota reservoir in 2018.

Waste Land: It was discovered that the amount of waste land in the area has decreased from 2007 to 2018. (Fig: 4). Waste land covered 34.41 sq km in 2007, accounting for 18.95 percent of total area, compared to 13.71 sq km in 2018, accounting for 7.55 percent of total area (Fig: 5). It's possible that the growth in waste land area from 2007 to 2018 is due to the conversion of waste land to agricultural land.

Mining: It was shown that from 2007 to 2018, area mining increased at a faster rate (Fig: 4). Mining covered 0.32 sq km in 2007, accounting for 0.17 percent of the total area, rising to 0.67 sq km in 2018, accounting for 0.369

percent of the entire area (Fig: 5). Mining area has increased from 2007 to 2018 may be due to continuous mining activates and mining leases for exploration.

Crop Land: It was discovered that from 2007 to 2018, the area crop increased at a faster rate (Fig: 4). Crop land covered 16.71 sq km in 2007, accounting for 9.20 percent of total area, compared to 35.98 sq km in 2018, accounting for 19.81 percent of total area (Fig: 5). The conversion of forest land, grassland, and woodland into crops resulted in an increase in mining area from 2007 to 2018.

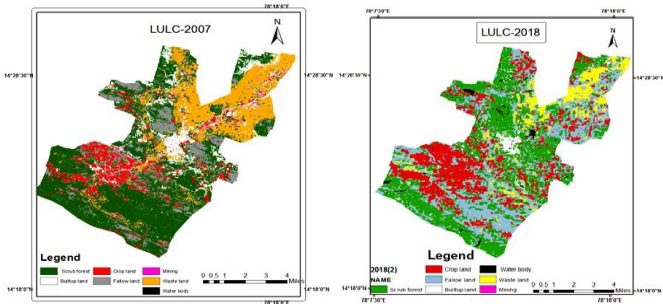


Fig. 4 Classified LULC Map of the Study Area in 2007 and 2018

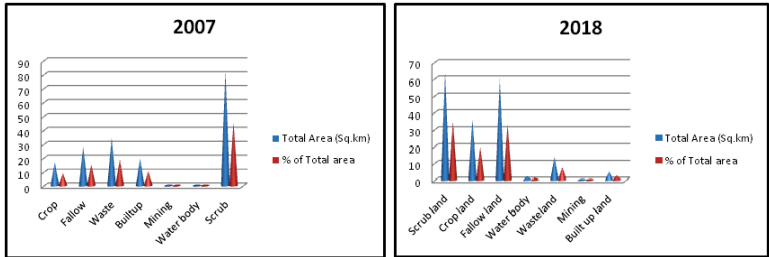


Fig. 5 Percentage of Variance in LULC Changes of the Study Area of the Year 2007 & 2018

5. Conclusion:

Using remote sensing data and GIS technologies, this research examines LU/LC variations in the Pulivendula mandal. Our findings clearly demonstrate that LU/LC variations were large from 2007 to 2018. There has been a huge increase in the built-up area. On the other

side, agricultural land, water spread land, and forest land are all shrinking. The impact of population and development activities on LU/LC change is clearly demonstrated in this study. This research demonstrates that combining GIS with remote sensing technology might be a useful tool for urban planning and management. This project will help environmental management groups, legislators, and the general public better comprehend the study area's surrounding environment.

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